

Supplementary Information for
*Climate-Informed Spatio-Temporal Modeling of COVID-19 Hotspot
Persistence in India (March–December 2020)*

Overview and Scope

This Supplementary Information (SI) accompanies the main manuscript *Climate-Informed Spatio-Temporal Modeling of COVID-19 Hotspot Persistence in India (March–December 2020)*. It provides full statistical and data appendices supporting the hierarchical binomial, GEE, and Markov analyses. All plots, figures, and schematic illustrations appear in the main text; the present document includes the complete state-wise tables and harmonized data panels.

Context. The COVID-19 pandemic demonstrated pronounced spatial and temporal heterogeneity across India [1–3]. State-wise hotspot proportions for March–December 2020 were compiled from the openly available dataset by Sharma *et al.* [?], who provided cluster interconnect-edness maps of COVID-19 spread in India via the Zenodo repository. The present SI reproduces the derived data tables used in the modeling pipeline described in the main manuscript.

Structure.

- §S1 — Data harmonization and denominators;
- §S2 — Full GEE coefficient tables (exchangeable and AR(1));
- §S3 — Mixture clustering diagnostics and memberships;
- §S4 — Markov transition counts and probabilities;
- §S5 — Declarations, data, and code availability.

S1 Data Harmonization and Denominators

State and Union Territory names were normalized (NCT OF DELHI→DELHI, UTTARANCHAL→UTTARAKHAND, PONDICHERRY→PUDUCHERRY) before merging with the district geolocation table. The merged panel yields a binomial denominator N_s (district count) for each state, ensuring coherent modeling of proportions as $Y_{st} \sim \text{Binomial}(N_s, p_{st})$.

The hotspot information originates from:

Arpan Sharma, Madhab Nirola, Prasanna Dhungel, & Mousumi Gupta. (2023).
Cluster Interconnectedness Maps of COVID-19 Spread in India (Version 2) [Data
set]. Zenodo. <https://doi.org/10.5281/zenodo.7981043>.

The following tables list the harmonized state-level denominators and monthly hotspot proportions.

Table S1: Overview By State

state	climate	N_s	$months_observed$	$mean_prop$
Andaman & Nicobar	Tropical Monsoon	3	10	0.0
Andhra Pradesh	Tropical Savannah, Arid, Steppe and Hot	13	10	0.078
Arunachal Pradesh	Montanne Climate and Humid Subtropical	26	10	0.002
Assam	Humid Subtropical	33	10	0.019
Bihar	Humid Subtropical	38	10	0.028
Chandigarh	Arid, Steppe and Hot	1	10	0.001
Delhi	Arid, Steppe and Hot	11	10	0.058
Goa	Tropical Monsoon	2	10	0.002
Gujarat	Arid, Steppe, Hot and Hot deserts	34	10	0.058
Haryana	Arid, Steppe and Hot	22	10	0.02
Himachal Pradesh	Montanne Climate and Humid Subtropical	12	10	0.001
Jammu and Kashmir	Montane Climate	20	10	0.006
Jharkhand	Humid Subtropical and Tropical Savannah	24	10	0.007
Karnataka	Tropical Savannah and Tropical Monsoon	30	10	0.048
Kerala	Tropical Monsoon	14	10	0.202
Ladakh	Montanne Climate	2	10	0.001
Lakshadweep	Tropical Monsoon	1	10	0.0
Madhya Pradesh	Tropical Savannah, Arid, Steppe, Hot and Humid Subtropical	52	10	0.014
Maharashtra	Tropical Savannah and Arid, Steppe, Hot	36	10	0.175
Manipur	Humid Subtropical	16	10	0.002
Meghalaya	Humid Subtropical	11	10	0.001
Mizoram	Humid Subtropical	11	10	0.0
Nagaland	Humid Subtropical	11	10	0.0
Odisha	Tropical Savannah	30	10	0.012
Puducherry	Tropical Savannah	4	10	0.003
Punjab	Arid, Steppe and Hot	22	10	0.017
Rajasthan	Hot Deserts, Arid	33	10	0.037
Sikkim	Montanne Climate	4	10	0.0
Telengana	Tropical Savannah, Arid, Steppe and Hot	33	10	0.018

state	climate	N_s	$months_observed$	$mean_prop$
Tripura	Humid Subtropical	8	10	0.001
Uttar Pradesh	Humid Subtropical and Arid, Steppe and Hot	75	10	0.034
Uttarakhand	Montane Climate and Humid Subtropical	13	10	0.004
West Bengal	Tropical Savannah and Humid Subtropical	23	10	0.026

Table S2: State Month Panel Long

state	state_norm	climate	N_s	$lat_centroid$
Andaman & Nicobar	ANDAMAN AND NICOBAR ISLANDS	Tropical Monsoon	3	10.50
Andhra Pradesh	ANDHRA PRADESH	Tropical Savannah, Arid, Steppe and Hot	13	16.08
Arunachal Pradesh	ARUNACHAL PRADESH	Montanne Climate and Humid Subtropical	26	27.85
Assam	ASSAM	Humid Subtropical	33	26.37
Bihar	BIHAR	Humid Subtropical	38	25.67
Chandigarh	CHANDIGARH	Arid, Steppe and Hot	1	30.73
Delhi	DELHI	Arid, Steppe and Hot	11	28.64
Goa	GOA	Tropical Monsoon	2	15.31
Gujarat	GUJARAT	Arid, Steppe, Hot and Hot deserts	34	22.38
Haryana	HARYANA	Arid, Steppe and Hot	22	29.13
Himachal Pradesh	HIMACHAL PRADESH	Montanne Climate and Humid Subtropical	12	31.63
Jammu and Kashmir	JAMMU AND KASHMIR	Montane Climate	20	33.54
Jharkhand	JHARKHAND	Humid Subtropical and Tropical Savannah	24	23.77
Karnataka	KARNATAKA	Tropical Savannah and Tropical Monsoon	30	14.37
Kerala	KERALA	Tropical Monsoon	14	10.37
Ladakh	LADAKH	Montanne Climate	2	34.35
Lakshadweep	LAKSHADWEEP	Tropical Monsoon	1	9.98
Madhya Pradesh	MADHYA PRADESH	Tropical Savannah, Arid, Steppe, Hot and Humid Subtropical	52	23.63
Maharashtra	MAHARASHTRA	Tropical Savannah and Arid, Steppe, Hot	36	19.43
Manipur	MANIPUR	Humid Subtropical	16	24.73
Meghalaya	MEGHALAYA	Humid Subtropical	11	25.54
Mizoram	MIZORAM	Humid Subtropical	11	23.36
Nagaland	NAGALAND	Humid Subtropical	11	26.05
Odisha	ODISHA	Tropical Savannah	30	20.51
Puducherry	PUDUCHERRY	Tropical Savannah	4	12.82
Punjab	PUNJAB	Arid, Steppe and Hot	22	30.92
Rajasthan	RAJASTHAN	Hot Deserts, Arid	33	26.23
Sikkim	SIKKIM	Montanne Climate	4	27.43
Telangana	TELANGANA	Tropical Savannah, Arid, Steppe and Hot	33	17.75

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Tripura	TRIPURA	Humid Subtropical	8	23.80
Uttar Pradesh	UTTAR PRADESH	Humid Subtropical and Arid, Steppe and Hot	75	27.02
Uttarakhand	UTTARAKHAND	Montane Climate and Humid Subtropical	13	29.88
West Bengal	WEST BENGAL	Tropical Savannah and Humid Subtropical	23	24.13
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West Bengal	WEST BENGAL	Tropical Savannah and Humid Subtropical	23	24.13
Andaman & Nicobar	ANDAMAN AND NICOBAR ISLANDS	Tropical Monsoon	3	10.50
Andhra Pradesh	ANDHRA PRADESH	Tropical Savannah, Arid, Steppe and Hot	13	16.08
Arunachal Pradesh	ARUNACHAL PRADESH	Montanne Climate and Humid Subtropical	26	27.85
Assam	ASSAM	Humid Subtropical	33	26.37
Bihar	BIHAR	Humid Subtropical	38	25.67
Chandigarh	CHANDIGARH	Arid, Steppe and Hot	1	30.73
Delhi	DELHI	Arid, Steppe and Hot	11	28.64
Goa	GOA	Tropical Monsoon	2	15.31
Gujarat	GUJARAT	Arid, Steppe, Hot and Hot deserts	34	22.38
Haryana	HARYANA	Arid, Steppe and Hot	22	29.13
Himachal Pradesh	HIMACHAL PRADESH	Montanne Climate and Humid Subtropical	12	31.63
Jammu and Kashmir	JAMMU AND KASHMIR	Montane Climate	20	33.54
Jharkhand	JHARKHAND	Humid Subtropical and Tropical Savannah	24	23.77

state	state_norm	climate	N_s	$lat_centroid$
Karnataka	KARNATAKA	Tropical Savannah and Tropical Monsoon	30	14.37
Kerala	KERALA	Tropical Monsoon	14	10.37
Ladakh	LADAKH	Montanne Climate	2	34.35
Lakshadweep	LAKSHADWEEP	Tropical Monsoon	1	9.98
Madhya Pradesh	MADHYA PRADESH	Tropical Savannah, Arid, Steppe, Hot and Humid Subtropical	52	23.63
Maharashtra	MAHARASHTRA	Tropical Savannah and Arid, Steppe, Hot	36	19.43
Manipur	MANIPUR	Humid Subtropical	16	24.73
Meghalaya	MEGHALAYA	Humid Subtropical	11	25.54
Mizoram	MIZORAM	Humid Subtropical	11	23.36
Nagaland	NAGALAND	Humid Subtropical	11	26.05
Odisha	ODISHA	Tropical Savannah	30	20.51
Puducherry	PUDUCHERRY	Tropical Savannah	4	12.82
Punjab	PUNJAB	Arid, Steppe and Hot	22	30.92
Rajasthan	RAJASTHAN	Hot Deserts, Arid	33	26.23
Sikkim	SIKKIM	Montanne Climate	4	27.43
Telangana	TELANGANA	Tropical Savannah, Arid, Steppe and Hot	33	17.75
Tripura	TRIPURA	Humid Subtropical	8	23.80
Uttar Pradesh	UTTAR PRADESH	Humid Subtropical and Arid, Steppe and Hot	75	27.02
Uttarakhand	UTTARAKHAND	Montane Climate and Humid Subtropical	13	29.88
West Bengal	WEST BENGAL	Tropical Savannah and Humid Subtropical	23	24.13
Andaman & Nicobar	ANDAMAN AND NICOBAR ISLANDS	Tropical Monsoon	3	10.50
Andhra Pradesh	ANDHRA PRADESH	Tropical Savannah, Arid, Steppe and Hot	13	16.08
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state	state_norm	climate	N_s	$lat_centroid$
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Ladakh	LADAKH	Montanne Climate	2	34.35
Lakshadweep	LAKSHADWEEP	Tropical Monsoon	1	9.98
Madhya Pradesh	MADHYA PRADESH	Tropical Savannah, Arid, Steppe, Hot and Humid Subtropical	52	23.63
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Tripura	TRIPURA	Humid Subtropical	8	23.80
Uttar Pradesh	UTTAR PRADESH	Humid Subtropical and Arid, Steppe and Hot	75	27.02
Uttarakhand	UTTARAKHAND	Montane Climate and Humid Subtropical	13	29.88
West Bengal	WEST BENGAL	Tropical Savannah and Humid Subtropical	23	24.13

Interpretation. N_s gives the number of districts per state used as the binomial denominator; **prop** is the monthly hotspot proportion; and $Y = \text{round}(N_s \times \text{prop})$ is the implied hotspot count used in the GLMM/GEE analyses.

S2 Population-Average Effects (GEE)

Full coefficients from the binomial GEE models (logistic link) are reported below. The exchangeable correlation is the primary specification; AR(1) serves as robustness. Each table lists estimates, robust standard errors, z statistics, and p values.

Table S3: Gee Exchangeable Coefficients

term	<i>coef</i>	<i>std_err</i>	<i>z</i>	<i>p</i>
const	−3.829	0.385	−9.935	0.0
climate _{Arid, Steppe, HotandHotdeserts}	0.858	0.301	2.846	0.004
climate _{HotDeserts, Arid}	0.364	0.301	1.208	0.227
climate _{HumidSubtropical}	−0.659	0.495	−1.331	0.183
climate _{HumidSubtropicalandArid, SteppeandHot}	0.269	0.301	0.893	0.372
climate _{HumidSubtropicalandTropicalSavannah}	−1.281	0.301	−4.252	0.0
climate _{MontaneClimate}	−1.424	0.301	−4.723	0.0
climate _{MontaneClimateandHumidSubtropical}	−2.011	0.301	−6.672	0.0
climate _{MontanneClimate}	−4.333	0.412	−10.518	0.0
climate _{MontanneClimateandHumidSubtropical}	−2.92	0.311	−9.393	0.0
climate _{TropicalMonsoon}	1.78	0.501	3.551	0.0
climate _{TropicalSavannah}	−0.848	0.325	−2.606	0.009
climate _{TropicalSavannahandArid, Steppe, Hot}	2.091	0.302	6.927	0.0
climate _{TropicalSavannahandHumidSubtropical}	−0.013	0.301	−0.043	0.966
climate _{TropicalSavannahandTropicalMonsoon}	0.649	0.301	2.152	0.031
climate _{TropicalSavannah, Arid, SteppeandHot}	0.31	0.589	0.526	0.599
climate _{TropicalSavannah, Arid, Steppe, HotandHumidSubtropical}	−0.625	0.301	−2.074	0.038
month _{cat₂020 − 04}	0.532	0.251	2.123	0.034
month _{cat₂020 − 05}	0.311	0.362	0.857	0.391
month _{cat₂020 − 06}	0.256	0.265	0.964	0.335
month _{cat₂020 − 07}	0.373	0.202	1.842	0.066
month _{cat₂020 − 08}	0.144	0.321	0.449	0.654
month _{cat₂020 − 09}	0.376	0.289	1.301	0.193
month _{cat₂020 − 10}	0.197	0.274	0.72	0.472
month _{cat₂020 − 11}	−0.193	0.879	−0.22	0.826
month _{cat₂020 − 12}	−0.189	0.89	−0.212	0.832

Table S4: Gee Ar1 Coefficients

term	<i>coef</i>	<i>std_err</i>	<i>z</i>	<i>p</i>
const	−3.989	0.383	−10.422	0.0
climate _{Arid, Steppe, HotandHotdeserts}	0.952	0.298	3.192	0.001
climate _{HotDeserts, Arid}	0.719	0.298	2.41	0.016
climate _{HumidSubtropical}	−0.76	0.508	−1.496	0.135
climate _{HumidSubtropicalandArid, SteppeandHot}	0.572	0.298	1.92	0.055
climate _{HumidSubtropicalandTropicalSavannah}	−1.206	0.298	−4.05	0.0
climate _{MontaneClimate}	−0.939	0.298	−3.152	0.002
climate _{MontaneClimateandHumidSubtropical}	−2.309	0.298	−7.75	0.0
climate _{MontanneClimate}	−4.071	0.566	−7.192	0.0
climate _{MontanneClimateandHumidSubtropical}	−2.615	0.316	−8.283	0.0
climate _{TropicalMonsoon}	2.113	0.5	4.224	0.0
climate _{TropicalSavannah}	−0.64	0.327	−1.959	0.05
climate _{TropicalSavannahandArid, Steppe, Hot}	2.267	0.302	7.518	0.0
climate _{TropicalSavannahandHumidSubtropical}	−0.118	0.298	−0.396	0.692
climate _{TropicalSavannahandTropicalMonsoon}	0.696	0.298	2.333	0.02
climate _{TropicalSavannah, Arid, SteppeandHot}	0.142	0.441	0.322	0.748
climate _{TropicalSavannah, Arid, Steppe, HotandHumidSubtropical}	−0.246	0.298	−0.825	0.409
month _{cat2020 − 04}	0.533	0.251	2.127	0.033
month _{cat2020 − 05}	0.312	0.365	0.856	0.392
month _{cat2020 − 06}	0.257	0.266	0.966	0.334
month _{cat2020 − 07}	0.374	0.204	1.833	0.067
month _{cat2020 − 08}	0.151	0.328	0.46	0.645
month _{cat2020 − 09}	0.384	0.294	1.305	0.192
month _{cat2020 − 10}	0.216	0.287	0.751	0.452
month _{cat2020 − 11}	−0.179	0.884	−0.202	0.84
month _{cat2020 − 12}	−0.179	0.893	−0.201	0.841

S3 Model-Based Clustering of Temporal Trajectories

This section summarizes the Gaussian mixture model selection and cluster memberships used to identify archetypal temporal hotspot trajectories.

Table S5: Gmm Bic

K	BIC
1.0	1348.411
2.0	1147.625
3.0	837.312
4.0	834.54
5.0	484.335

Table S6: State Clusters

state	cluster
Andaman & Nicobar	3
Andhra Pradesh	1
Arunachal Pradesh	3
Assam	2
Bihar	2
Chandigarh	3
Delhi	1
Goa	0
Gujarat	4
Haryana	4
Himachal Pradesh	0
Jammu and Kashmir	0
Jharkhand	0
Karnataka	1
Kerala	2
Ladakh	3
Lakshadweep	3
Madhya Pradesh	2
Maharashtra	1

state	cluster
Manipur	0
Meghalaya	3
Mizoram	3
Nagaland	3
Odisha	0
Puducherry	0
Punjab	2
Rajasthan	1
Sikkim	3
Telangana	2
Tripura	3
Uttar Pradesh	1
Uttarakhand	0
West Bengal	2

Table S7: Cluster Mean Trajectories

cluster	2020 – 03	2020 – 04	2020 – 05	2020 – 06	2020 – 07	2020 – 08	2020 – 09
0.0	0.0	0.003	0.003	0.003	0.006	0.007	0.007
1.0	0.084	0.098	0.077	0.081	0.082	0.098	0.098
2.0	0.022	0.018	0.031	0.02	0.034	0.024	0.024
3.0	0.0	0.0	0.0	0.0	0.0	0.001	0.001
4.0	0.0	0.092	0.109	0.068	0.043	0.018	0.018

Note. Diagnostic plots (BIC curve, per-cluster trajectories, and climate-wise distributions) are shown in the main manuscript (Figures 2–5).

S4 Markov Transitions for Hotspot Status

Two-state transitions summarize persistence and relapse of hotspot status ($Z_{st} = \mathbb{1}\{Y_{st} \geq 1\}$). π_{11} denotes persistence, and π_{01} denotes relapse.

Table S8: Markov Transition Counts

climate	Z_now		n
Arid, Steppe and Hot	0	0	15
Arid, Steppe and Hot	0	1	6
Arid, Steppe and Hot	1	0	8
Arid, Steppe and Hot	1	1	7
Arid, Steppe, Hot and Hot deserts	0	0	2
Arid, Steppe, Hot and Hot deserts	0	1	1
Arid, Steppe, Hot and Hot deserts	1	0	1
Arid, Steppe, Hot and Hot deserts	1	1	5
Hot Deserts, Arid	1	0	1
Hot Deserts, Arid	1	1	8
Humid Subtropical	0	0	48
Humid Subtropical	0	1	4
Humid Subtropical	1	0	5
Humid Subtropical	1	1	6
Humid Subtropical and Arid, Steppe and Hot	1	1	9
Humid Subtropical and Tropical Savannah	0	0	7
Humid Subtropical and Tropical Savannah	0	1	1
Humid Subtropical and Tropical Savannah	1	0	1
Montane Climate	0	0	9
Montane Climate and Humid Subtropical	0	0	9
Montane Climate	0	0	18
Montane Climate and Humid Subtropical	0	0	18
Tropical Monsoon	0	0	33
Tropical Monsoon	0	1	1
Tropical Monsoon	1	1	2
Tropical Savannah	0	0	14
Tropical Savannah	0	1	1
Tropical Savannah	1	0	1
Tropical Savannah	1	1	2

climate		Z_now	n
Tropical Savannah and Arid, Steppe, Hot	1	1	9
Tropical Savannah and Humid Subtropical	0	0	2
Tropical Savannah and Humid Subtropical	0	1	1
Tropical Savannah and Humid Subtropical	1	0	1
Tropical Savannah and Humid Subtropical	1	1	5
Tropical Savannah and Tropical Monsoon	0	0	1
Tropical Savannah and Tropical Monsoon	1	0	1
Tropical Savannah and Tropical Monsoon	1	1	7
Tropical Savannah, Arid, Steppe and Hot	0	0	6
Tropical Savannah, Arid, Steppe and Hot	0	1	1
Tropical Savannah, Arid, Steppe and Hot	1	0	3
Tropical Savannah, Arid, Steppe and Hot	1	1	8
Tropical Savannah, Arid, Steppe, Hot and Humid Subtropical	0	0	2
Tropical Savannah, Arid, Steppe, Hot and Humid Subtropical	0	1	3
Tropical Savannah, Arid, Steppe, Hot and Humid Subtropical	1	0	2
Tropical Savannah, Arid, Steppe, Hot and Humid Subtropical	1	1	2

Table S9: Markov Transition Probabilities

climate	i	j	n		p_ij
Arid, Steppe and Hot	0	0	15	21	0.7143
Arid, Steppe and Hot	0	1	6	21	0.2857
Arid, Steppe and Hot	1	0	8	15	0.5333
Arid, Steppe and Hot	1	1	7	15	0.4667
Arid, Steppe, Hot and Hot deserts	0	0	2	3	0.6667
Arid, Steppe, Hot and Hot deserts	0	1	1	3	0.3333
Arid, Steppe, Hot and Hot deserts	1	0	1	6	0.1667
Arid, Steppe, Hot and Hot deserts	1	1	5	6	0.8333
Hot Deserts, Arid	0	0	0	0	
Hot Deserts, Arid	0	1	0	0	

climate	i	j	n	p_ij	
Hot Deserts, Arid	1	0	1	9	0.1111
Hot Deserts, Arid	1	1	8	9	0.8889
Humid Subtropical	0	0	48	52	0.9231
Humid Subtropical	0	1	4	52	0.0769
Humid Subtropical	1	0	5	11	0.4545
Humid Subtropical	1	1	6	11	0.5455
Humid Subtropical and Arid, Steppe and Hot	0	0	0	0	
Humid Subtropical and Arid, Steppe and Hot	0	1	0	0	
Humid Subtropical and Arid, Steppe and Hot	1	0	0	9	0.0
Humid Subtropical and Arid, Steppe and Hot	1	1	9	9	1.0
Humid Subtropical and Tropical Savannah	0	0	7	8	0.875
Humid Subtropical and Tropical Savannah	0	1	1	8	0.125
Humid Subtropical and Tropical Savannah	1	0	1	1	1.0
Humid Subtropical and Tropical Savannah	1	1	0	1	0.0
Montane Climate	0	0	9	9	1.0
Montane Climate	0	1	0	9	0.0
Montane Climate	1	0	0	0	
Montane Climate	1	1	0	0	
Montane Climate and Humid Subtropical	0	0	9	9	1.0
Montane Climate and Humid Subtropical	0	1	0	9	0.0
Montane Climate and Humid Subtropical	1	0	0	0	
Montane Climate and Humid Subtropical	1	1	0	0	
Montanne Climate	0	0	18	18	1.0
Montanne Climate	0	1	0	18	0.0
Montanne Climate	1	0	0	0	
Montanne Climate	1	1	0	0	
Montanne Climate and Humid Subtropical	0	0	18	18	1.0
Montanne Climate and Humid Subtropical	0	1	0	18	0.0
Montanne Climate and Humid Subtropical	1	0	0	0	
Montanne Climate and Humid Subtropical	1	1	0	0	
Tropical Monsoon	0	0	33	34	0.9706

climate	i	j	n		p_ij
Tropical Monsoon	0	1	1	34	0.0294
Tropical Monsoon	1	0	0	2	0.0
Tropical Monsoon	1	1	2	2	1.0
Tropical Savannah	0	0	14	15	0.9333
Tropical Savannah	0	1	1	15	0.0667
Tropical Savannah	1	0	1	3	0.3333
Tropical Savannah	1	1	2	3	0.6667
Tropical Savannah and Arid, Steppe, Hot	0	0	0	0	
Tropical Savannah and Arid, Steppe, Hot	0	1	0	0	
Tropical Savannah and Arid, Steppe, Hot	1	0	0	9	0.0
Tropical Savannah and Arid, Steppe, Hot	1	1	9	9	1.0
Tropical Savannah and Humid Subtropical	0	0	2	3	0.6667
Tropical Savannah and Humid Subtropical	0	1	1	3	0.3333
Tropical Savannah and Humid Subtropical	1	0	1	6	0.1667
Tropical Savannah and Humid Subtropical	1	1	5	6	0.8333
Tropical Savannah and Tropical Monsoon	0	0	1	1	1.0
Tropical Savannah and Tropical Monsoon	0	1	0	1	0.0
Tropical Savannah and Tropical Monsoon	1	0	1	8	0.125
Tropical Savannah and Tropical Monsoon	1	1	7	8	0.875
Tropical Savannah, Arid, Steppe and Hot	0	0	6	7	0.8571
Tropical Savannah, Arid, Steppe and Hot	0	1	1	7	0.1429
Tropical Savannah, Arid, Steppe and Hot	1	0	3	11	0.2727
Tropical Savannah, Arid, Steppe and Hot	1	1	8	11	0.7273
Tropical Savannah, Arid, Steppe, Hot and Humid Subtropical	0	0	2	5	0.4
Tropical Savannah, Arid, Steppe, Hot and Humid Subtropical	0	1	3	5	0.6
Tropical Savannah, Arid, Steppe, Hot and Humid Subtropical	1	0	2	4	0.5
Tropical Savannah, Arid, Steppe, Hot and Humid Subtropical	1	1	2	4	0.5

Interpretation. Persistence π_{11} is higher in tropical and humid climates; relapse π_{01} is generally low. Denominators (see transition counts) inform the reliability of these probabilities. Associated maps and time-series illustrations appear in the main paper.

S5 Declarations and Availability

Code availability. All analysis scripts (Python 3.12, pandas, statsmodels, geopandas) that reproduce the SI tables are available upon publication under a CC-BY 4.0 license (Zenodo snapshot).

Data availability. We employ the open dataset of Sharma *et al.* (2023) [?], *Cluster Interconnectedness Maps of COVID-19 Spread in India (Version 2)* [Zenodo dataset, <https://doi.org/10.5281/zenodo.7981043>].

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